

THE AUTONOMOUS AI CAREER OPERATING SYSTEM

Transforming Redrob into an AI-Native SaaS Ecosystem

Architectural Presentation Deck & Strategy Overview

Outlining What We Built, Why We Built It That Way, and How It Works.

Core Stack: Next.js 15 • React 19 • FastAPI • LangGraph • Qdrant Vector DB • Neo4j Graph DB • Redis

1. WHAT WE BUILT: 12 Autonomous AI Agents

Redrob delegates continuous career execution to specialized autonomous AI agents working 24x7:

Executive Agent

Orchestrates global workflows, assigns goals, maintains memory.

Career Coach Agent

Predicts trajectory, salary benchmarks, and 12-month roadmaps.

Resume & Skill Gap Agent

ATS scoring, keyword optimization, and missing skill detection.

Job Discovery Agent

Scans thousands of jobs and auto-saves matched opportunities.

Interview Agent

Simulates voice, coding, and behavioral interviews with feedback.

Recruiter Suite Agent

Multi-signal candidate ranking and hiring analytics.

Networking & Learning Agent

Curates courses, YouTube videos, and alumni mentor graphs.

24x7 Opportunity Monitor

Autonomous daemon checking hackathons, freelance & jobs.

2. WHY WE BUILT IT THAT WAY: Architectural Rationale

Strategic architectural decisions driving platform scalability and intelligence:

Architectural Decision	Rationale & Strategic Value
AI-First Autonomous Execution	Replaces manual job application fatigue with proactive 24x7 background background execution by specialized agents.
Multi-Provider LLM Failover	Unified provider factory routes calls dynamically across OpenAI, Gemini, Claude, Groq, and local Ollama without vendor lock-in.
Hybrid DB Architecture	Combines PostgreSQL (transactional records), Qdrant (vector semantic resume matching), and Neo4j (skill dependency graphs).
Next.js 15 Glassmorphism UI	Delivers an ultra-premium, dark-mode visual interface with real-time WebSockets telemetry and interactive agent widgets.

3. HOW IT WORKS: End-to-End Execution Flow

Lifecycle execution from user onboarding to autonomous candidate placement:

- 1. Profile Ingestion & Parsing:** User uploads resume → Parsed into structured JSON and embedded into Qdrant vector database.
- 2. Knowledge Graph Mapping:** Skills mapped into Neo4j graph nodes to detect missing proficiencies and project gaps.
- 3. Multi-Agent Collaboration:** LangGraph coordinates state execution between Executive, Career Coach, and Resume agents.
- 4. Real-time Mock Interview Lab:** Candidate practices voice/behavioral questions → Receives instant sentiment and clarity feedback.
- 5. Recruiter Candidate Ranking:** Recruiter runs semantic queries → Multi-signal engine computes weighted composite fit scores.
- 6. 24x7 Continuous Monitoring:** Background daemon scans market opportunities and streams real-time notifications via WebSockets.

Production Readiness Status: 100% Code Complete • Docker Orchestrated • 100% Pytest Suite Success